

TSENAT: Tsallis Entropy Analysis Toolbox

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Overview

TSENAT (Tsallis Entropy Analysis Toolbox) quantifies **isoform heterogeneity** in RNA-seq data using Tsallis entropy—a scale-dependent diversity measure distinct from abundance-focused tools (DESeq2, Salmon) and differential transcript usage packages (*SplicingFactory*). By tuning a sensitivity parameter q , users examine diversity at different scales: rare variants (low q) or dominant isoforms (high q). This package enables computing Tsallis entropy from transcript-level abundance estimates, comparing measures between groups, and visualizing scale-dependent differences via q -curves. Results integrate seamlessly with Bioconductor’s `SummarizedExperiment`, capturing scale-dependent isoform complexity that often signals cellular state changes independent of total abundance shifts.

Motivation and Bioconductor Contribution

Common RNA-seq tools focus on either total gene abundance changes or individual transcript usage shifts. Yet genes often remodel their isoform landscapes without changing overall expression—a phenomenon missed by these approaches. TSENAT quantifies this *isoform complexity* directly via Tsallis entropy, a scale-dependent diversity framework tuned by parameter q . By sliding q across scales, researchers zoom between rare variants (low q) and dominant isoforms (high q), capturing biological signal invisible to abundance- or proportion-based summaries.

In this guide, we demonstrate the complete workflow: preprocessing transcript counts, computing entropy across q -values, testing for between-group differences, and visualizing scale-dependent complexity. All results integrate seamlessly with Bioconductor’s `SummarizedExperiment`, making TSENAT a natural complement to existing DTU and abundance-focused tools in the ecosystem.

High-level workflow

1. **Load and preprocess** transcript counts using `build_analysis_s4()`, then filter low-abundance transcripts with `filter_analysis_s4()`.
2. **Select q -values** tuned to your research question: low q (rare isoforms) vs. high q (dominant isoforms). Compute Tsallis entropy across these scales with `calculate_diversity_s4()`.
3. **Test for differences** in entropy between groups using `rank_test_q_condition_s4()` (Wilcoxon or permutation tests) (Kerby 2014; Saulsbury 2020).
4. **Visualize results** as q -curves and inspect transcript counts for genes with the strongest entropy shifts using `plot_tsallis_q_curve_s4()` and related functions.

Design assumptions: This guide assumes paired or longitudinal designs with 6–8 samples per group for adequate power to detect entropy shifts while controlling false discovery rate. Smaller sample sizes may be underpowered to detect subtle isoform complexity changes.

Installation

To install TSENAT from Bioconductor:

```
if (!require("BiocManager", quietly = TRUE))
  install.packages("BiocManager", repos = "http://cran.us.r-project.org")
BiocManager::install("TSENAT")
```

Then load the package:

```
library(TSENAT)
```

Quick Start

For users who want to see results quickly, here is a minimal workflow:

```
library(TSENAT)
library(SummarizedExperiment)

# Load example data
data("readcounts", package = "TSENAT")
readcounts <- as.matrix(readcounts)

# Load sample metadata and annotation
metadata_df <- read.table(
  system.file("extdata", "metadata.tsv", package = "TSENAT"),
  header = TRUE, sep = "\t"
)
gff3_file <- system.file("extdata", "annotation.gff3.gz", package = "TSENAT")

# Create analysis object with transcript counts, annotation, and metadata
analysis <- build_analysis_s4(readcounts, gff3_file, metadata = metadata_df)

# Filter low-abundance transcripts
analysis <- filter_analysis_s4(analysis, stringency = "medium")

# Compute Tsallis entropy using S4 wrapper (using single q value for quick start)
analysis <- calculate_diversity_s4(analysis, q = 1, norm = TRUE)
```

Data Structure Overview

Input Data: TSENAT expects transcript-level read counts (rows = transcripts, columns = samples) from quantification tools like SALMON or kallisto. A GFF3 annotation file maps transcripts to genes.

Processing: The `build_analysis_s4()` function accepts these inputs and creates a **TSENATAnalysis** S4 object, which is the central data container throughout your analysis.

Organization: The TSENATAnalysis object encapsulates:

- `@se` — SummarizedExperiment storing counts and metadata (access via `se(analysis)`)
- `@diversity_results` — Entropy values across q-values (accessor: `diversity(analysis)`)
- `@jackknife_results` — Jackknife confidence intervals and isoform switching results

- `@lm_results` — LM interaction statistics and rank test q-value effects (accessor: `lmResults(analysis)`)
- `@divergence_results` — Pairwise divergence metrics (accessor: `divergence(analysis)`)
- `@plots` — Generated visualizations

Use **accessor functions** like `diversity()`, `divergence()`, `lmResults()`, etc. for type-safe result extraction without direct slot manipulation.

For more details on the `SummarizedExperiment` class, see `SummarizedExperiment` documentation.

What is Entropy and Tsallis Entropy?

The Problem: Measuring Isoform Complexity

Standard RNA-seq analysis measures *whether* transcript abundance changes between conditions. But a critical complementary question remains underexplored: *how* does the diversity of isoforms change? A gene may show little change in total abundance while dramatically reshuffling its isoform repertoire—a phenomenon that current methods largely miss.

Entropy quantifies precisely this: the complexity, richness, and balance of isoform heterogeneity. By measuring entropy across different biological scales, researchers can detect whether changes are driven by shifts in rare variants or reorganization of dominant isoforms.

Tsallis Entropy: A Scale-Dependent Diversity Measure

Tsallis entropy (Tsallis 2006) is a one-parameter family of diversity measures that generalizes Shannon entropy (Shannon 1948). Unlike Shannon entropy (which treats all isoforms equally), Tsallis entropy lets you tune a sensitivity parameter q to zoom into different aspects of isoform complexity:

- $q < 1$: Emphasizes rare, low-abundance isoforms (discovery mode)
- $q = 1$: Recovers Shannon entropy (balanced across all abundance scales)
- $q = 2$: Emphasizes dominant isoforms (robustness mode)
- $q > 2$: Focuses almost exclusively on the most abundant species

This is the key innovation: by computing entropy across a range of q -values (a “ q -curve”), you obtain a complete picture of isoform heterogeneity (Rényi 1961). In TSENAT, you’ll explore multiple q -values using `calculate_diversity_s4()` and visualize results with `plot_tsallis_q_curve_s4()`.

Mathematical Definition

For a discrete probability vector $p = (p_1, \dots, p_n)$ representing isoform proportions within a gene, Tsallis entropy is:

$$S_q(p) = \frac{1 - \sum_{i=1}^n p_i^q}{q - 1}.$$

This parametric family unites diverse entropy concepts under a single framework:

- **Generalization**: Extends beyond Shannon entropy to capture scale-dependent phenomena
- **Mathematical elegance**: Reduces to well-known diversity indices at specific q -values
- **Practical flexibility**: Enables data-driven exploration across the full diversity spectrum

Information-Theoretic Interpretation

From an information theory perspective (Shannon 1948; Tsallis 2006), entropy measures the uncertainty or surprise when drawing a single transcript from an isoform distribution. Higher entropy means the draw is less predictable (many similarly abundant isoforms), while lower entropy means one or a few isoforms dominate. This distinction is crucial: two genes with identical total abundance may have dramatically different isoform complexity.

The q parameter acts as a **sensitivity dial**. Consider a simple example: - A gene with 5 equally abundant isoforms: all q -values give high entropy - A gene where 1 isoform dominates: low- q entropy stays high (rare variants matter), while high- q entropy drops (rare variants ignored)

This scale-dependent nature reveals biological signal invisible to abundance- or proportion-based summaries alone.

Special Cases and Limiting Behavior

Tsallis entropy exhibits important special cases (Masi 2005) that appear when you set specific q -values:

- **$q = 0$ (Richness):** $S_0 = m - 1$ (number of expressed isoforms). Pure species count, most minimal assumption.
- **$q \rightarrow 1$ (Shannon):** $S_1(p) = -\sum_i p_i \log p_i$. Standard information entropy (Shannon 1948); optimal estimator of information content.
- **$q = 2$ (Gini-Simpson):** $S_2 = 1 - \sum_i p_i^2$. Probability that two randomly drawn transcripts differ (Simpson 1949); robust to rare variants.

Normalized entropy: For genes with different isoform counts, normalize by maximum possible entropy: $\tilde{S}_q = \frac{S_q(p)}{S_{q,\max}(m)}$ where $S_{q,\max}(m) = \frac{1-m^{1-q}}{q-1}$. This enables fair comparison across genes.

Biological Potentiality: Why TSENAT Matters

Tsallis entropy is grounded in Shannon’s foundational information theory (Shannon 1948), generalized by Rényi’s family of entropies (Rényi 1961), and extended by Tsallis (Tsallis 2006). The key innovation is a tunable parameter q that controls sensitivity to different aspects of the diversity distribution—which aspects of isoform heterogeneity become visible depends on the q -value chosen (Anastasiadis 2012; Ramírez-Reyes et al. 2016; Alomani and Kayid 2023).

As stated explicitly in the mathematical literature (Masi 2005): “This introduces the formal possibility not to set rare and common events on the same footing, as in BG or Shannon statistics, but it enhances or depresses them according to the parameter chosen.” This principle is formalized in the Hill numbers framework (Chao et al. 2010), which unifies diverse entropy-based measures (richness, Shannon, Simpson) as different manifestations of the same parametric family with varying sensitivity to abundance scales.

The flexibility is crucial for isoform analysis. The concept of “true diversity” (Chao et al. 2010) emphasizes that diversity can be decomposed into two independent components: species richness (how many distinct isoforms exist) and evenness (how evenly distributed they are across the population). Two genes can have identical Shannon entropy yet differ dramatically in isoform structure: one might be dominated by a single abundant isoform (maximizing apparent heterogeneity at low q where rare variants matter), while another distributes transcripts across many isoforms equally (maximizing heterogeneity across all q values). By tuning the q -parameter, we operationalize what Masi (2005) describes: depressing the weight on rare events (high q) reveals dominant isoform patterns, while enhancing rare-event weight (low q) reveals cryptic isoform complexity. TSENAT’s multi- q approach captures this full spectrum of isoform diversity, enabling researchers to detect both rare isoform innovations and robust isoform usage patterns (Drost 2018) by exploring the q -curve across multiple scales.

Potential Applications in Genomics

The multi-scale nature of Tsallis entropy makes it suited for exploring isoform complexity across diverse biological contexts. By measuring information content at different q -values, researchers can detect patterns invisible to traditional transcript abundance measures alone:

Isoform complexity as a biological signal: Isoform switching—reorganization of the isoform landscape without necessarily changing total gene abundance—reflects strategic shifts in protein function driven by splicing regulation. Evidence from single-cell transcriptomics demonstrates that transcript-level complexity varies systematically across cell types and developmental states (Cao et al. 2017), validating that isoform heterogeneity is a genuine biological phenomenon rather than noise. Increased entropy in gene regulatory networks drives phenotypic heterogeneity and cellular plasticity (Nijman 2020), suggesting that transcript-level entropy captures similar organizational principles. TSENAT enables detection of these changes through entropy-based approaches, which capture whether complexity is increasing (diversity spreading across isoforms) or decreasing (consolidation onto dominant isoforms).

Scale-dependent organization: Different biological processes may prioritize different scales of organization (Tarabichi et al. 2013): - Changes in rare isoform usage (revealed through low q sensitivity) might reflect exploratory or error-correction mechanisms - Shifts in dominant isoform selection (revealed through high q sensitivity) might reflect functional specialization or robustness demands - The full q -curve reveals whether cellular transitions involve wholesale reorganization or targeted adjustments, paralleling how systems biology approaches dissect biological disorder and organization at multiple scales

Beyond abundance measures: Traditional analysis focuses on fold-changes and differential abundance. Since isoform reorganization can occur independently of total abundance changes, entropy-based approaches complement classical methods by detecting complexity shifts that transcript-level statistics alone cannot reveal.

Evidence from Literature The recent emphasis on information-theoretic approaches in computational biology Bajić (2024) reflects broader recognition that complex biological systems encode information across multiple organizational scales. Peer-reviewed literature provides strong empirical support for these applications:

Entropy and cancer biology: Cancer cells accumulate genetic and epigenetic perturbations that systematically increase disorder in gene regulatory networks. As Tarabichi and colleagues demonstrate, “Increased entropy of signaling (or gene interaction networks) has been well studied as a cancer characteristic: Network entropy increases along with cancer progresses” (Tarabichi et al. 2013). Systems biology approaches reveal that this entropy increase, rather than being an incidental feature, actively drives cancer progression through selection of cells with greater network flexibility and adaptability.

Perturbation-driven mechanisms of heterogeneity: Nijman’s analysis of perturbation-driven entropy proposes a complementary mechanism: “cancer-associated perturbations collectively disrupt normal gene regulatory networks by increasing their entropy. Importantly, in this model both somatic driver and passenger alterations contribute to ‘perturbation-driven entropy’, thereby increasing phenotypic heterogeneity and evolvability” (Nijman 2020). This framework elegantly explains observed cancer heterogeneity without requiring that every genetic change confers a selective advantage—some mutations contribute entropy directly through network disruption.

Single-cell validation of transcript diversity: Cao and colleagues’ landmark single-cell transcriptomics study provided empirical validation that transcript-level organization varies systematically across cell types: “expression levels of mRNA species are linked to cellular function and therefore can be used to classify cell types” (Cao et al. 2017). Their comprehensive profiling of *C. elegans* demonstrates that individual cells maintain specific, consistent isoform compositions reflecting cellular identity—establishing that isoform complexity is not noise but a fundamental aspect of cellular differentiation and function.

These three perspectives—entropy as driver of cancer evolution, perturbations as network disruption mechanisms, and single-cell evidence for systematic isoform organization—converge on a framework where measur-

ing entropy at multiple scales (via TSENAT's multi- q approach) captures biologically meaningful variation in cellular organization and adaptation. In cancer biology specifically, entropy-driven mechanisms explain how perturbations increase heterogeneity and plasticity (Nijman 2020), demonstrating that entropy concepts have mechanistic relevance beyond abstract information theory. Tsallis entropy, through the parametric q -spectrum, provides a systematic framework for exploring this multi-scale organization at the transcript and isoform level.

Isoform Switching Workflow

We now demonstrate a complete workflow for detecting **isoform switching**—when cells reorganize their isoform landscape without necessarily changing total gene abundance. This often reflects strategic shifts in protein function driven by splicing regulation.

In this workflow, you will:

1. Load transcript counts and sample metadata
2. Identify genes with significant scale-dependent isoform complexity changes (using linear model interaction testing)
3. Assess which individual transcripts drive those changes (using jackknife resampling (Efron and Tibshirani 1993))
4. Interpret results in the context of paired experimental designs

We'll work with a paired experimental design where treated and control samples are linked, enabling detection of robust biological signals while controlling for subject-level variability. By the end, you'll know not just *which genes* undergo isoform switching, but *which transcripts* are responsible and *at which diversity scales* the switching occurs.

Load data and metadata

An example dataset is included for demonstration.

```
# Load packages
suppressPackageStartupMessages({
  library(TSENAT)
  library(ggplot2)
  library(SummarizedExperiment)
})

# Setup random seed for reproducibility
set.seed(42)
```

Now we will load the example dataset and associated metadata:

```
# Load example dataset (lazy-loaded as 'readcounts' by default)
# This includes SALMON preprocessing outputs:
# - readcounts: transcript-level NumReads (raw fragment counts)
# - tpm: TPM matrix (length- and library-normalized estimates)
# - effective_length: EffectiveLength vector (read-length corrected)
data(readcounts)

readcounts <- as.matrix(readcounts)
```

```

metadata_df <- read.table(
  system.file("extdata", "metadata.tsv", package = "TSENAT"),
  header = TRUE, sep = "\t"
)

gff3_file <- system.file("extdata", "annotation.gff3.gz", package = "TSENAT")

```

Data preprocessing and filtering

Next, we will create a `TSENATAnalysis` object containing a `SummarizedExperiment` data container (a standard Bioconductor class for storing high-throughput assay data along with associated metadata). You can read more about it in the Bioconductor documentation.

We will use the GFF3.gz annotation file for extracting the transcript-to-gene mapping and building the analysis object. We'll also include sample metadata for improved analysis. Following Bioconductor best practices, we create the configuration `FIRST`, then pass it to the builder function (fail-fast principle):

```

## Configure analysis parameters first (best practice: fail-fast principle)
## This validates all parameters against metadata before object creation
config <- tsenat_config(
  condition_col = "condition",
  subject_col = "paired_samples",
  q_values = seq(0, 2, by = 0.05),
  nthreads = 2,
  paired = TRUE,
  control = "normal",
  metadata = metadata_df
)

## Build a complete `TSENATAnalysis` object from readcounts + GFF3.gz annotation
## Pass config at construction (Bioconductor pattern): immutable object creation
analysis <- build_analysis_s4(
  config = config,
  readcounts = readcounts,
  tx2gene = gff3_file,
  tpm = tpm,
  effective_length = effective_length
)

```

The `TSENATAnalysis` object contains a `SummarizedExperiment` with transcript-level counts and gene annotations extracted from the GFF3 file.

To reduce noise and improve statistical power, we filter out lowly-expressed transcripts. Expression estimates of transcript isoforms with zero or low expression might be highly variable (Jose and Lal 2013; Chakraborty 2019). For more details on the effect of transcript isoform prefiltering on differential transcript usage, see this paper.

```

analysis <- filter_analysis_s4(analysis, stringency = "medium")
cat("Retained", nrow(se(analysis)), "transcripts after filtering\n")
#> Retained 341 transcripts after filtering

```

The `stringency = "medium"` parameter keeps transcripts present in 50% of samples with TPM values above the median of mean gene TPM. This balances noise reduction with preservation of isoform diversity needed for meaningful entropy calculations.

Compute Tsallis Entropy

We now compute Tsallis entropy across your configured q-spectrum. Diversity measures provide a comprehensive framework for assessing isoform heterogeneity at multiple scales (Ramírez-Reyes et al. 2016; Gao et al. 2019). We normalize entropy to [0,1] range (`norm = TRUE`) for comparable cross-q assessment.

```
# Compute diversity using S4 wrapper with bootstrap confidence intervals [@S232; @S115]
analysis <- calculate_diversity_s4(
  analysis,
  norm = TRUE
)
#> Note: 12 genes excluded (< 75% valid values).
```

```
#> Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be
#> resized.
```

Table 1: **Table 1:** Tsallis entropy for first 4 genes across three diversity scales (sample: SRR14800481)

Gene	q=0.0	q=0.5	q=1.0	q=1.5	q=2.0
FOXJ2	0.66667	0.53796	0.48150	0.46975	0.48518
TMEM38A	1.00000	0.82704	0.72526	0.66965	0.64401
CCNE1	0.33333	0.32256	0.32749	0.34571	0.37418
MCOLN1	1.00000	0.39297	0.21567	0.16305	0.15195

With the q-spectrum we can produce a q-curve per sample and gene. These curves show how diversity emphasis shifts from rare to dominant isoforms as q increases and form the basis for interaction tests. The q-curve shows entropy as a function of q . Diverging curves between groups indicate scale-dependent diversity differences: separation at low q implies differences in rare isoforms, while separation at high q signals differences in dominant isoforms.

```
# Plot overall q-curve
p_qcurve <- plot_tsallis_q_curve_s4(
  se = analysis
)
print(p_qcurve)
```

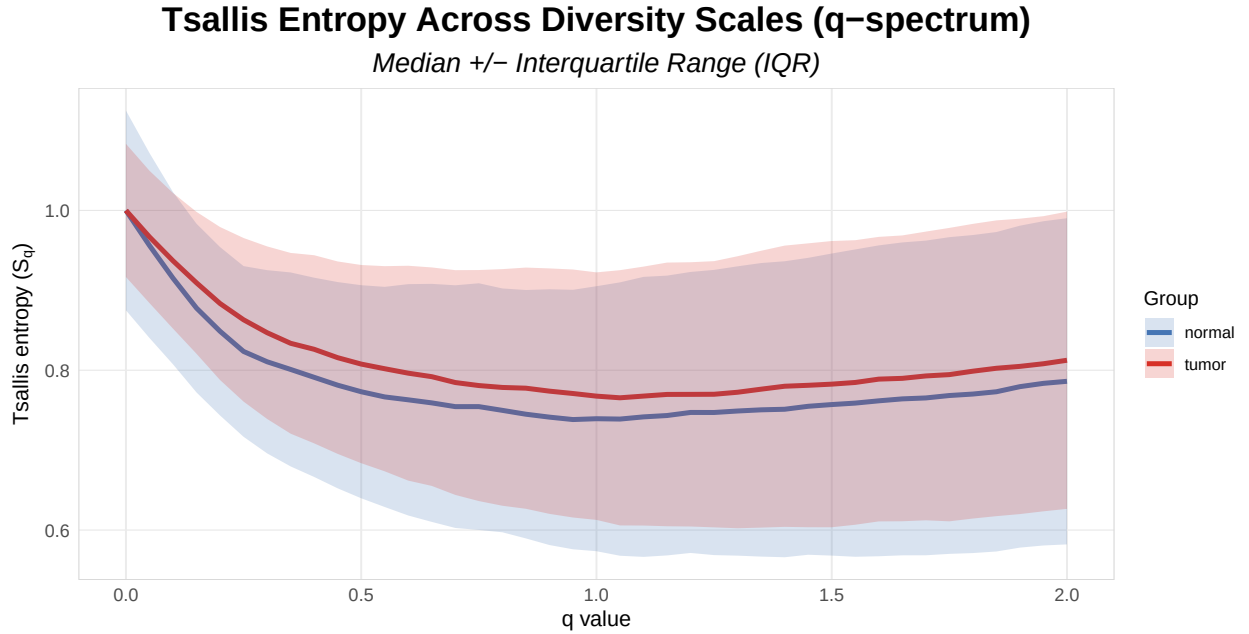



Figure 1: **Figure 1:** Isoform diversity profiles across q-values. Lines show normalized Tsallis entropy (0–1) for each sample, blue control and red treatment.

Quality Control: Sample Influence Assessment

Before proceeding with group-level comparisons, we should assess whether individual samples exert disproportionate influence on our entropy estimates (Efron, Bradley and Tibshirani, Robert J. 1993). To address this, we employ a leave-one-out influence assessment combined with robust M-estimation (Phipson and Smyth 2010) (iteratively re-weighted least squares with Huber loss). This approach quantifies how much each sample's removal affects the estimated location differences across the q-spectrum, providing a sample-level quality control metric independent of group assignment.

Samples with high influence scores exert disproportionate leverage on entropy estimates and may warrant deeper investigation for technical quality issues, subject-specific outliers, or genuine biological signals that require careful interpretation.

We configure three key parameters: `loss_type = "huber"` employs Huber M-estimation (robust to outliers with bounded influence), `q_combine_method = "mean"` averages influence metrics across the entire q-spectrum for a single score per sample, and `influence_threshold = 0.75` designates the percentile cut-off above which samples are flagged (such that the top 25% most influential samples are reviewed).

```
# Perform multi-q sample influence analysis via S4 wrapper
analysis <- m_estimate_s4(
  analysis,
  loss_type = "huber",
  q_combine_method = "mean",
  influence_threshold = 0.75
)

# Extract results from metadata using accessor function
sample_qc <- metadata(analysis, "m_estimate_results")
```

```
#> Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be
```

```
#> resized.
```

Table 2: ****Table 2 | Sample Influence Assessment via M-estimation.**** Ranked by magnitude of resampling-based influence values. Columns: Sample identifier; mean and standard deviation of Tsallis entropy; distance from centroid; outlier status. M-estimation identifies influential samples for sensitivity analysis.

Sample	Condition	Proportion_Affected	Genes_Affected	Entropy_Mean	Entropy_SD	Distance_from_
SRR14800481	normal	0.847	64.4	0.6936	0.2364	
SRR14800480	normal	0.806	61.2	0.7345	0.1972	
SRR14800477	normal	0.833	63.3	0.6896	0.2852	
SRR14800476	normal	0.889	67.6	0.7310	0.2329	
SRR14800475	normal	0.861	65.4	0.6774	0.2435	
SRR14800490	normal	0.849	64.5	0.6843	0.2357	
SRR14800489	normal	0.875	66.5	0.7339	0.2426	
SRR14800488	normal	0.861	65.4	0.6738	0.2649	
SRR14800479	tumor	0.861	65.4	0.6842	0.2721	
SRR14800478	tumor	0.903	68.6	0.7345	0.2440	
SRR14800487	tumor	0.861	65.4	0.7257	0.2193	
SRR14800486	tumor	0.903	68.6	0.7207	0.2497	
SRR14800485	tumor	0.861	65.4	0.7334	0.2131	
SRR14800484	tumor	0.889	67.6	0.7382	0.2095	
SRR14800483	tumor	0.861	65.4	0.7425	0.2283	
SRR14800482	tumor	0.917	69.7	0.7186	0.2446	

Interpretation: Samples with distance from centroid >1.5 or proportion of affected genes >0.85 are flagged for quality control review. Flagged samples may reflect genuine biological heterogeneity or technical artifacts requiring careful examination before proceeding with downstream analysis.

Linear-Model interaction

We can test for interactions between q and sample groups across the q -sequence using one of five methods: linear, lmm (linear mixed model (Le Roux and Rouanet 2011; Phipson and Smyth 2010)), gam (generalized additive model (Chapman 2017; Goude 2024)), gee (generalized estimating equations for paired data), or fpca (functional principal component analysis). The gam method flexibly captures nonlinear q -response patterns and is particularly useful for complex interaction structures.

```
# Linear-model interaction test across q values using S4 wrapper
if (requireNamespace("mgcv", quietly = TRUE)) {
  analysis <- calculate_lm_interaction_s4(
    analysis,
    method = "gam",
    multicorr = "hochberg"
  )
}
```

```
#> Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be
#> resized.
```

Table 3: ****Table 3 | Leading genes with scale-dependent condition effects from generalized additive models.**** We display the 6 genes with lowest adjusted p-values ($q^* < 0.05$). Benjamini-Hochberg correction applied; columns show gene identifier, effect size, test statistic, convergence status, and heteroscedasticity detection. Methods: GAMs with q^* and condition as smooth predictors [S190].

Gene	Gene Name	P-value	Adj. P-value	Effect Size	Test Statistic	Model Converged	Heteroscedasticity
CXCL12	CXCL12	3.93e-164	2.99e-162	0.8127	752.5086	TRUE	TRUE
THY1	THY1	9.80e-97	7.35e-95	0.6008	442.1369	TRUE	TRUE
ING3	ING3	2.72e-77	2.02e-75	0.3533	352.5945	TRUE	TRUE
SNHG10	SNHG10	9.81e-75	7.16e-73	0.0515	340.8200	TRUE	TRUE
LINC03040	LINC03040	1.87e-57	1.35e-55	0.7380	261.2380	TRUE	TRUE
HDAC2	HDAC2	2.61e-51	1.85e-49	0.2844	232.9431	TRUE	TRUE

Interpretation: TRUE in the Heteroscedasticity column indicates that the model satisfies homogeneity-of-variance assumptions across the q -spectrum; FALSE values suggest variance heterogeneity and warrant caution in result interpretation.

Now we will plot the q -curve profile for the top genes identified by the linear-model interaction test.

```
# Plot q-curve profiles for the top 4 genes using the S4 wrapper
combined_plot <- plot_lm_interaction_gam_s4(
  analysis,
  n_top = 4
)
print(combined_plot)
```

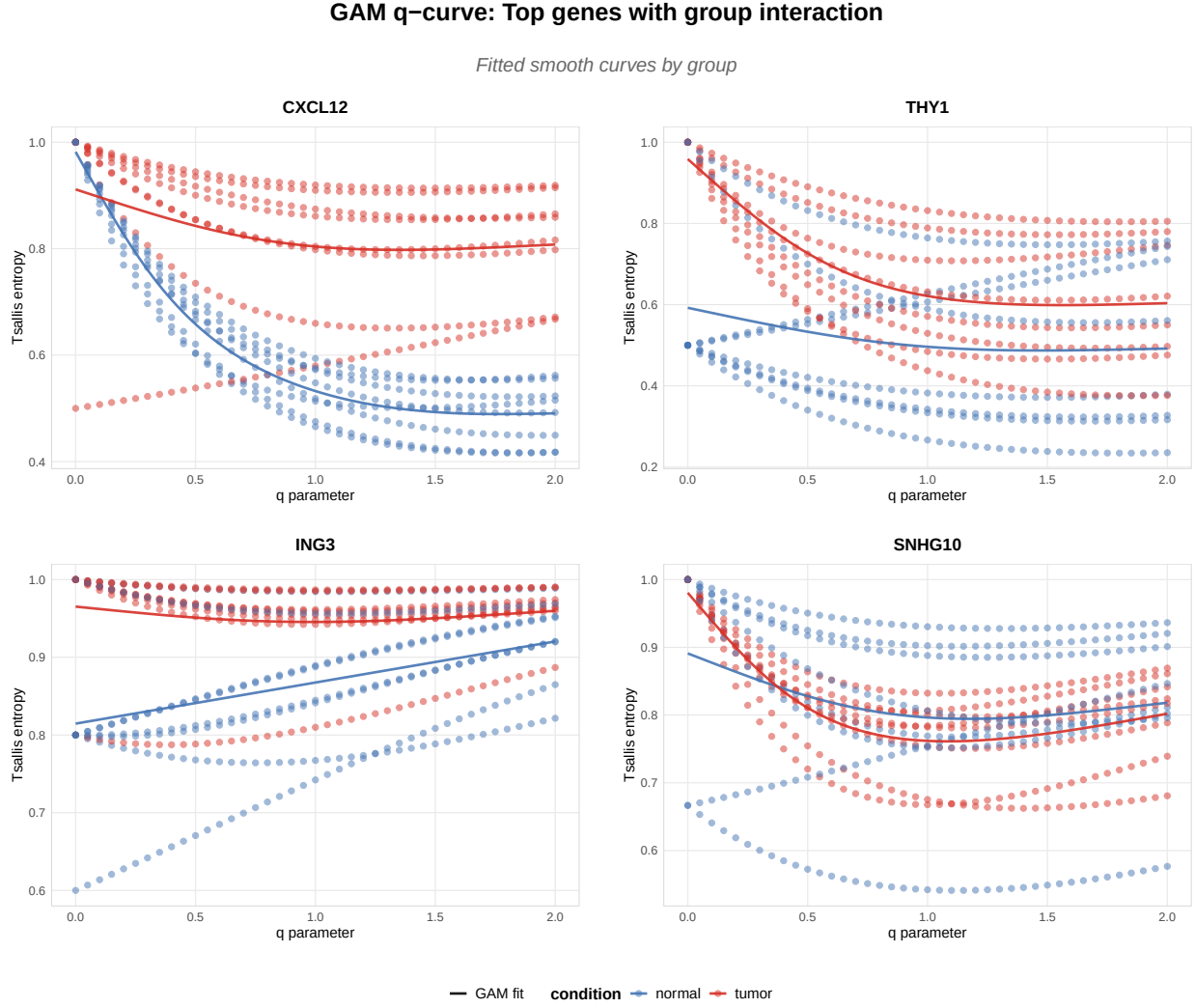


Figure 2: **Figure 2:** Scale-dependent interaction analysis. GAM-identified genes showing significant $q \times \text{condition}$ effects (Benjamini–Hochberg $q < 0.05$).

Transcript Switching Across Diversity Scales

Why Scale-Dependent Analysis Matters

Biological systems operate through processes at fundamentally different organizational scales. A single gene’s expression can be viewed through different “lenses”—from the perspective of dominant isoforms (high q , emphasizing abundant variants) to rare isoforms that may carry functional significance in specific contexts (low q , emphasizing rare variants). The LM interaction test revealed that condition effects vary across these diversity scales (q parameter), reflecting how biological reorganization can manifest differently depending on which aspect of isoform complexity we examine.

The theoretical foundation for scale-dependent analysis comes from generalized entropy measures. Rényi’s entropy family (Rényi 1961) introduced a continuous spectrum of diversity measures parameterized by a sensitivity index, which Tsallis later extended and popularized (Alomani and Kayid 2023). Unlike Shannon entropy (which represents a single point on this spectrum), Tsallis entropy provides a family of measures where the parameter q acts as a “magnifying glass”—adjusting what organizational scales become visible. Mathematically, different q -values weight rare versus abundant isoforms differently: - Low q (0-0.5): Em-

phasized rare isoforms, revealing exploratory or conditional splicing programs - Mid q (0.5-1.5): Balanced sensitivity across the full isoform distribution

- High q (1.5-2.0): Emphasized abundant isoforms, revealing core splicing decisions

In biological systems, these different scales often correspond to different regulatory mechanisms. A transcript may be consistently rare across conditions (no switching at low q) while being stably abundant at high q . Detecting such scale-specific switching patterns reveals the mechanistic structure of splicing regulation (Tarabichi et al. 2013).

Bridging Theory: Multi- q Analysis to Fine-Scale Jackknife Testing

The current analysis bridges two complementary approaches:

1. **Multi- q LM interaction test:** Tests whether the condition effect on divergence depends on q (overall pattern across diversity scales)
2. **Single- q jackknife resampling:** Tests which individual transcripts drive those patterns at each q -value, with robustness assessment

This two-stage approach combines hypothesis testing with robustness validation (Efron, Bradley and Tibshirani, Robert J. 1993). The LM test identifies *where* (at which q -values) significant switching occurs. The jackknife-based delta influence then identifies *which transcripts* contribute to those patterns, weighting confidence by operational stability rather than simple frequency.

Interpreting Scale-Dependent Switching Patterns

Here we assess whether these switching patterns are consistent across different q values or vary by scale. The key question is: **Do the same transcripts switch across all diversity scales, or does the identity of switching transcripts depend on which scale we examine?**

- **Consistent switching:** The same transcripts show significant delta influence across q -values $\rightarrow$ suggests a robust, scale-independent splicing shift
- **Mixed/scale-dependent switching:** Different transcripts show importance at different q -values $\rightarrow$ suggests regulatory complexity where the mechanism differs by scale

This classification reveals whether isoform reorganization is a coordinated, global phenomenon (same transcripts across scales) or a layered process where different regulatory inputs dominate at different organizational scales.

```
# Multi-Q analysis: Test switching at different q values
analysis <- jackknife_isoform_switching_s4(
  analysis,
  q = c(0, 0.5, 1, 1.5, 2),
  nboot = 100,
  lm_p_threshold = 0.05,
  threshold = 90
)
```

We configure three key parameters for the switching analysis: `nboot = 100` specifies 100 bootstrap resamples to estimate robust confidence intervals for delta influence values (Efron, Bradley and Tibshirani, Robert J. 1993); `lm_p_threshold = 0.05` filters genes to those with significant $q \times$ condition interaction effects ($p < 0.05$) before assessing transcript switching; and `threshold = 90` designates transcripts with support $\geq 90\%$

across bootstrap samples as robust switches. These parameters balance sensitivity (detecting switching) with specificity (avoiding false positives from noise).

The tables below examine transcript switching patterns for the top genes with strongest $q \times \text{condition}$ interactions. Delta influence, derived from jackknife resampling (Efron, Bradley and Tibshirani, Robert J. 1993), quantifies how each transcript's relative importance changes between conditions, weighted by stability across bootstrap iterations (positive = more influential in first condition; negative = second condition). The **Direction Consistency** column classifies each transcript: "Consistent" indicates stable switching across q -values, while "Mixed" indicates scale-dependent switching. This reveals whether isoform shifts involve the same transcripts across scales or whether different q -values emphasize different transcripts.

```
# Prepare comparison tables for top genes across q-values using S4 accessor
tables_result <- prepare_gene_switching_tables_s4(
  analysis,
  n_top_genes = 2
)
```

Gene: CXCL12 (ENSG00000107562.18)

```
#> Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be
#> resized.
```

Transcript	q=0.00	q=0.50	q=1.00	q=1.50	q=2.00	Direction Consistency
ENST00000343575.11	-0.033	0.000	0.041	0.062	0.073	Mixed directions
ENST00000374426.6	0.046	-0.063	-0.051	-0.040	-0.035	Mixed directions
ENST00000374429.6	-0.033	0.164	0.156	0.123	0.112	Mixed directions

Gene: THY1 (ENSG00000154096.15)

```
#> Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be
#> resized.
```

Transcript	q=0.00	q=0.50	q=1.00	q=1.50	q=2.00	Direction Consistency
ENST00000524970.5	-0.198	-0.046	0.002	0.014	0.020	Mixed directions
ENST00000900758.1	0.196	0.045	0.031	0.019	0.016	Consistent positive
ENST00000956364.1	0.118	-0.075	-0.092	-0.102	-0.108	Mixed directions

Delta Influence Across Diversity Scales

Visualize switching patterns for the top genes identified by the LM interaction test. This shows which transcripts are switching in genes with significant $q \times \text{condition}$ interaction effects. The heatmaps below display **jackknife delta influence** (Efron and Tibshirani 1993)—a resampling-based measure of how robustly each transcript's relative contribution changes between conditions—across different q -values (rows: 0 to 2.0, representing increasing diversity scales from rare to abundant isoforms) and transcripts (columns) for each gene. Delta influence values are computed from bootstrap resampling iterations, providing robust, non-parametric estimates of transcript switching significance weighted by consistency across replicates.

```
# Generate multi-Q heatmaps for top 4 genes using S4 wrapper
plot_multiq_delta_influence_heatmaps_s4(
  analysis,
  n_genes = 4
)
```

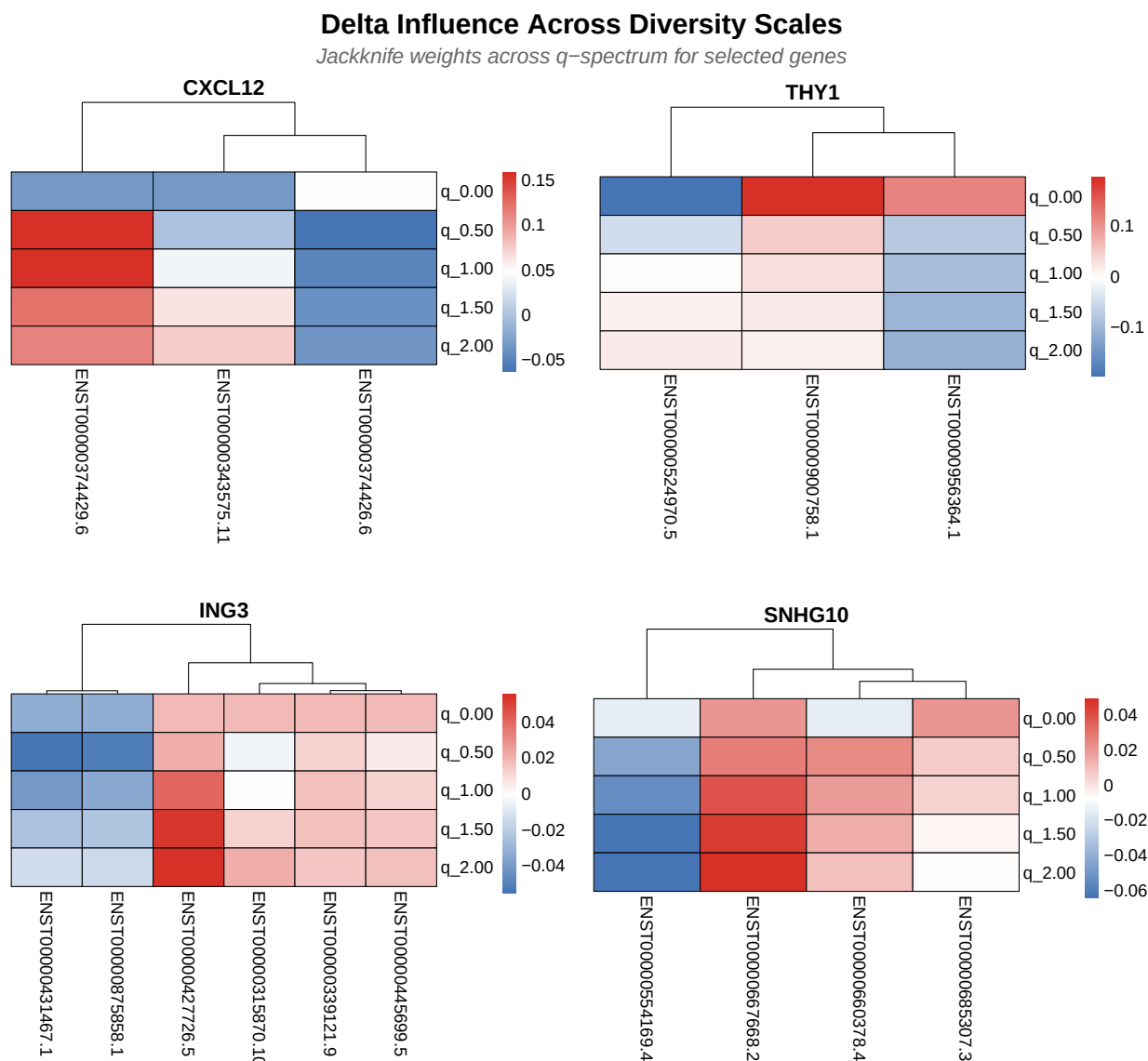


Figure 3: **Figure 3:** Transcript-level switching patterns via jackknife resampling. Rows q-values (0–2.0), columns transcripts. Red/blue indicates condition influence.

Interpretation: Rows represent q-values across the diversity spectrum (0 to 2.0, with low q emphasizing rare isoforms and high q emphasizing abundant isoforms). Red cells indicate transcripts with strong positive delta influence in the first condition (meaning their relative importance increased); blue indicates negative influence in the second condition (meaning their importance decreased). **Patterns to look for:**

- **Strong color intensity (bright red or blue):** Robust switching signal, consistently observed across bootstrap resamples
- **Weak color intensity (pale/white):** Uncertain or noisy switching signal, high variability across resamples
- **Consistent colors down a column:** That transcript switches stably across all q-values (scale-independent switching)
- **Varied colors down a column:** That transcript shows scale-dependent switching (different roles at rare vs. abundant scales)

- **Color differences across rows:** Different q-values emphasize different transcripts, revealing how regulatory mechanisms vary by topological scale

This robustness-weighted visualization reveals not just *which* transcripts switch, but *how reliably* they switch, allowing distinction between robust biological signals and artifacts of sampling variation.

```
# Generate transcript abundance heatmap for top 4 transcripts using hierarchical clustering
# metric = "median" aggregates expression values across paired conditions
plot_top_transcripts_s4(
  analysis,
  top_n = 4,
  metric = "median"
)
```

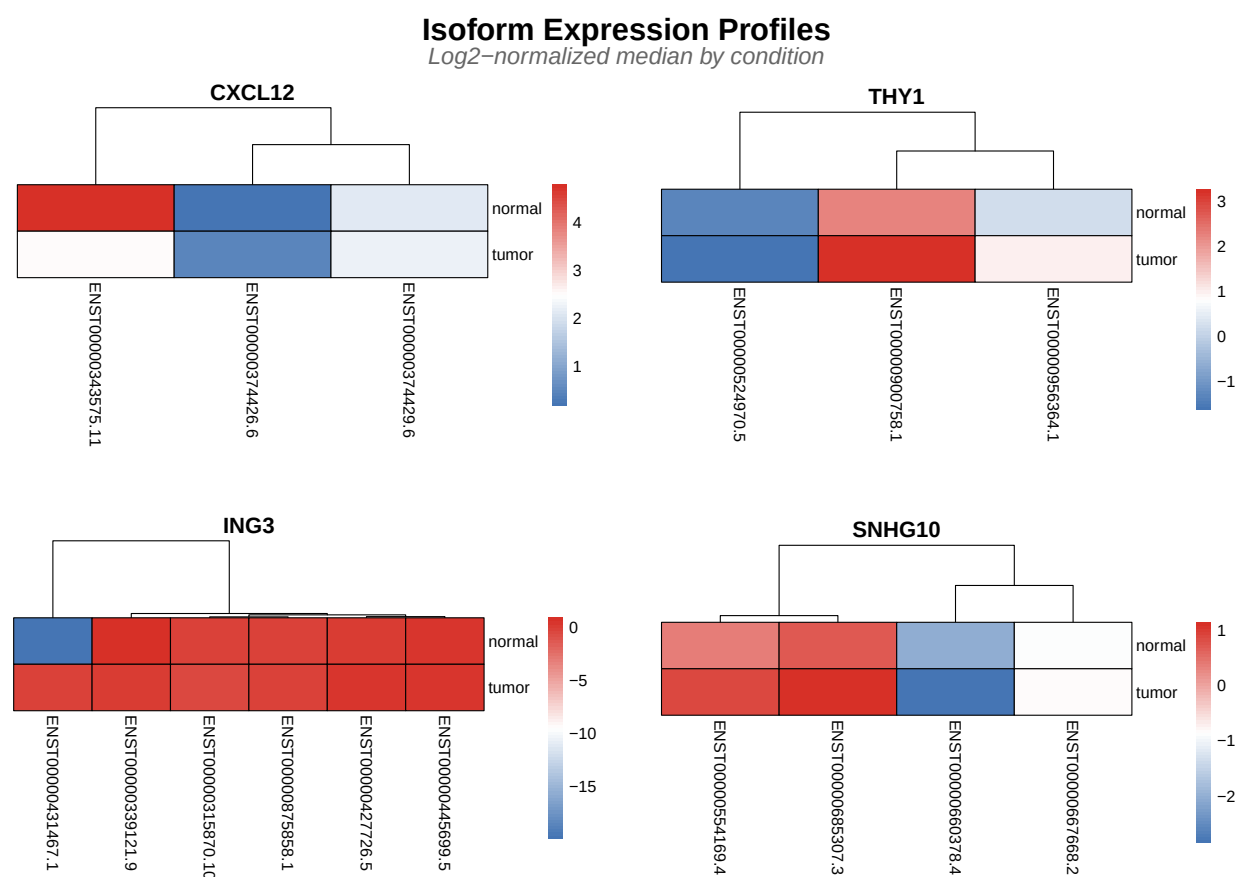


Figure 4: **Figure 4:** Transcript abundance heatmap with hierarchical clustering. Rows are transcripts, columns are conditions. Blue low, red high expression.

Interpretation: Transcripts (rows) are hierarchically clustered by expression similarity. Red indicates high expression, blue indicates low expression. Distinct horizontal bands reveal condition-specific isoforms; uniform coloring suggests constitutive expression. Clustering reveals which transcripts co-vary in expression patterns across conditions.

Effect size analysis

To understand which diversity scales show the most pronounced biological differences between groups, we compute effect size metrics (Tsallis divergence D_q) across the q-spectrum while respecting the paired sample design. This reveals whether group differences are driven by rare isoforms (low q), dominant isoforms (high q), or uniformly across scales.

The approach uses linear mixed-effects regression (LMM) with q as continuous predictor to assess slope differences in entropy change across q-spectrum between treatment groups, with effect size computed as Tsallis divergence $D_q(\text{control}||\text{treatment})$ (Kullback and Leibler 1951; Tsallis 2006; Rényi 1961; Sason 2022b, 2022a). This information-theoretic measure automatically respects Tsallis entropy properties and adapts to each q value (Shiner et al. 2002).

Theoretical Justification

Tsallis entropy theory Sason (2022b) demonstrates that different q values reveal fundamentally different aspects of the isoform distribution. As documented in the foundational divergence literature (Sason 2022b, 2022a; Ré and Azad 2014; R00e9 and Azad 2014), $D_{0.5}(P||Q)$ often differs substantially from $D_2(P||Q)$, with q=0.5 emphasizing rare variants and q=2 emphasizing abundant isoforms. Computing effect sizes independently at each q value thus captures which diversity scales (rare vs. dominant isoforms) show the strongest biological differentiation between conditions (Sfetcu et al. 2022; R00e9 and Azad 2014). Ré and Azad (2014) demonstrated that Tsallis divergence generalizations improve discrimination of genomic sequences, validating the multi-scale divergence approach for biological applications (R00e9 and Azad 2014).

Effect Size Interpretation

Tsallis divergence D_q quantifies the information-theoretic distance between two distributions' entropy patterns across the q-spectrum (Yulmetyev et al. 2004a; Shiner et al. 2007; Wang et al. 2021; Chernyshov 2009; Sason 2022a). For paired designs, divergence is computed separately for each pair and then averaged, accounting for within-pair correlation and reducing noise from between-pair variation (Yulmetyev et al. 2004b). The mean divergence across q values serves as the effect size, automatically capturing how entropy distributions differ between control and treatment groups at all scales (rare to abundant isoforms). Values $D > 0.1$ indicate meaningful divergence, following effect size classification thresholds (Kerby 2014; Chao et al. 2010; Sason 2022a). Genes with $D > 0.1$ demonstrate significant information-theoretic separation between conditions, revealing that isoform complexity patterns (captured by multi-q Tsallis entropy) fundamentally differ between treatment groups across all q-dependent scales (R00e9 and Azad 2014).

```
# Calculate divergence: computes pairwise information-theoretic distance (Tsallis divergence) between c  
# using bootstrap confidence intervals across all configured q-values from getConfig(analysis)$q_values  
analysis <- calculate_divergence_s4(  
  analysis)
```

```
# Compute effect sizes: significance_threshold = 0.05 filters genes to those with qxcondition interacti  
# enrich_per_q_pattern = TRUE classifies each gene by its divergence pattern (Rare-driven/Balanced/Abun  
analysis <- effect_sizes_divergence_s4(  
  analysis,  
  significance_threshold = 0.05,  
  enrich_per_q_pattern = TRUE  
)
```

```
#> Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be  
#> resized.
```

Table 6: **Table 5 | Top genes by linear model significance with effect sizes and q-spectrum patterns.** Ranked by statistical significance (ascending *P*-values, Benjamini–Hochberg *q*-value < 0.05). Columns: gene identifier; effect size (mean divergence across q-spectrum); pattern classification (Rare-driven/Balanced/Abundant-driven); pattern metrics—D_rare: median divergence for rare isoforms (q<1), D_abundant: median divergence for abundant isoforms (q>1), Ratio: D_rare/D_abundant (>1.3 indicates rare-driven, <0.77 indicates abundant-driven); adjusted P-value for linear model interaction test.

Gene	Mean Divergence	Q-Pattern	D_rare	D_abundant	Ratio	LM adj. p-value
CXCL12	0.2846	Balanced	0.3063	0.2712	1.13	3.0e-162
THY1	0.1279	Balanced	0.1455	0.1142	1.27	7.3e-95
ING3	0.0101	Rare driven	0.0120	0.0085	1.42	2.0e-75
SNHG10	0.0714	Rare driven	0.0861	0.0591	1.46	7.2e-73
LINC03040	0.2398	Rare driven	0.2813	0.2057	1.37	1.3e-55
HDAC2	0.0525	Rare driven	0.0642	0.0425	1.51	1.9e-49
ENSG00000274322	0.1506	Rare driven	0.1852	0.1206	1.54	3.3e-46
MEF2A	0.0298	Rare driven	0.0382	0.0225	1.70	9.5e-42
RAP1GDS1	0.0139	Rare driven	0.0168	0.0114	1.47	5.2e-36
PDE7A	0.0419	Rare driven	0.0490	0.0360	1.36	1.1e-35

Interpretation of Pattern Types:

Pattern Type	Signature	Biological Meaning
Rare driven	$D(q=0.5) \gg D(q=2.0)$	Low-abundance isoforms are condition-specific; treatment group preferentially expresses rare transcripts not seen in control
Abundant driver	$D(q=0.5) \ll D(q=2.0)$	High-abundance isoforms shift between conditions; treatment remodels the dominant transcript landscape without rare variants changing
Balanced	$D(q=0.5) \approx D(q=2.0)$	All isoforms shift proportionally; no preferential weighting to rare or abundant forms; suggests systematic rebalancing

The effect sizes reveal which genes show the most information-theoretic separation across the q-spectrum **between paired treatment groups**. Genes with Tsallis divergence $D > 0.1$ show substantial divergence, indicating that entropy distributions differ fundamentally between conditions across the full q-spectrum.

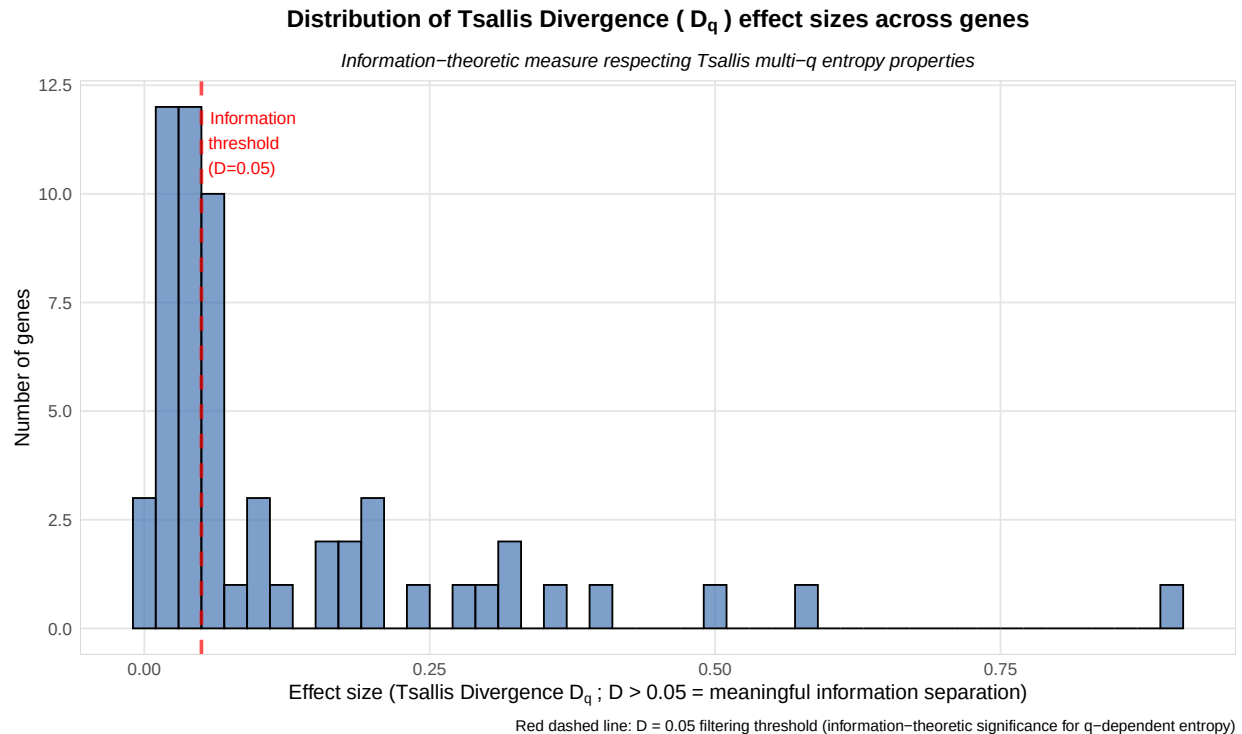
Biological Interpretation: Following Tsallis entropy theory and information-theoretic principles (Tsallis 2006; Rényi 1961; Sason 2022b; Hyndman and Athanasopoulos 2018), effect size quantification (Unknown 2020) genes with large D values are those where isoform complexity distributions differ qualitatively between conditions when examined at all sensitivity scales (q = rare → abundant isoforms). For example, a gene might show high entropy at low q (emphasizing rare isoforms) in one condition but low entropy at high q (emphasizing abundant isoforms) in another, revealing scale-dependent regulatory mechanisms. These genes are candidates for investigation of condition-specific splicing architecture and dynamic isoform switching.

```

# Visualize the distribution of Tsallis divergence effect sizes across genes
# The red dashed line marks  $D = 0.1$  (information-theoretic significance threshold)
# Generate divergence distribution plot (returns plot object)
# Optionally save to file for external use
plot_obj <- plot_divergence_distribution_s4(
  analysis,
  threshold = 0.05
)

print(plot_obj)

```



```

# Visualize  $q$ -spectrum curves for top 4 genes by significance
# Plot is rendered directly by knitr (no file dependency)
p_multi <- plot_divergence_spectrum_s4(
  analysis,
  n_genes = 4,
  use_pvalue_ranking = TRUE
)

print(p_multi)

```

Interpreting the Q -Spectrum Curve:

- Shape matters more than single value: The **q -spectrum** curve is the complete biological story. A flat curve (balanced) vs. declining curve (rare driven) vs. rising curve (abundant driven) encode fundamentally different mechanisms.
- $q=1$ (KL divergence): The middle point corresponds to ordinary Kullback-Leibler divergence (Kullback and Leibler 1951; R00e9 and Azad 2014), where in the limit $q \rightarrow 1$ the Tsallis divergence reduces to

Divergence Spectra: Per-gene Comparisons

Ranked by interaction significance (median)

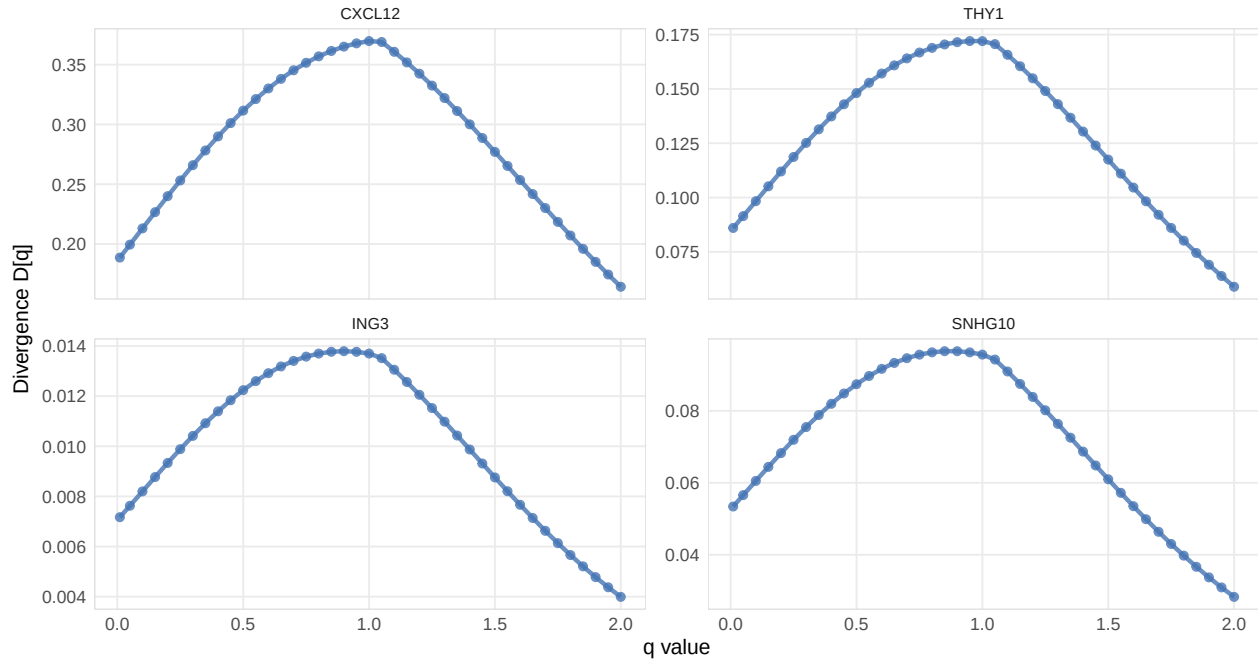


Figure 5: **Figure 5:** Q-spectrum curves for top genes by significance. Each line represents divergence as a function of q-value; shape reveals biological pattern (rare-driven, balanced, or abundant-driven).

standard KL divergence (R00e9 and Azad 2014). This weights all isoforms equally. This is the “average” effect.

- $q < 1$ (rare isoforms): Emphasizes how much low-abundance transcripts differ between groups. If $D(q=0.5) \gg D(q=1)$, the divergence is driven by changes in rare variants.
- $q > 1$ (abundant isoforms): Emphasizes dominant transcripts. If $D(q=2) \gg D(q=1)$, major isoforms rebalance while rare ones stay similar.
- Bootstrap CI bands: Shaded region shows 95% confidence limits on per-q estimates. Narrow bands indicate robust estimates; wide bands suggest noisy data or small sample sizes.

We can also plot the global divergence spectrum across all genes. This curve is the full biological signature of **isoform switching**—the pattern encodes which abundance scales (rare vs. abundant) are affected.

```
# Visualize global divergence q-spectrum across all genes (aggregated)
# Plot is rendered directly by knitr (no file dependency)
p <- plot_divergence_spectrum_s4(
  analysis
)

print(p)
```

Statistical Validation During Interpretation:

1. Bootstrap CI validity: Per-q CIs should narrow as q increases (Efron, Bradley and Tibshirani, Robert J. 1993) (higher q = aggregation effect, less variance). If CIs grow with q, check for data quality issues.

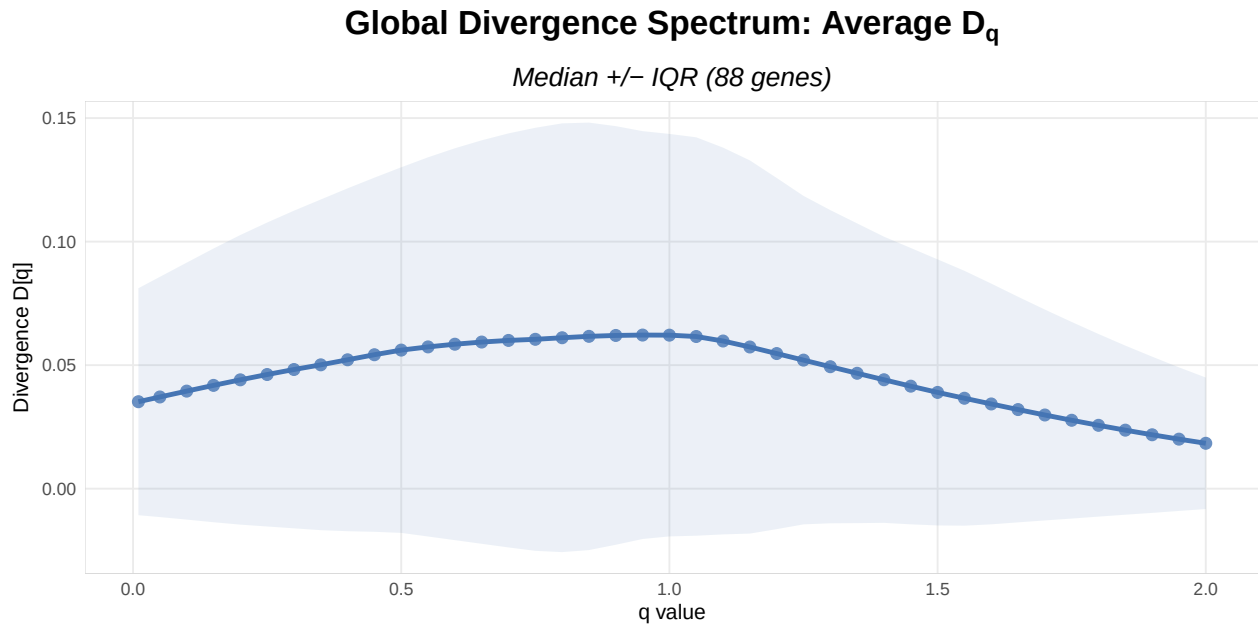


Figure 6: **Figure 6:** Global divergence spectrum across all genes. Aggregated q-spectrum pattern shows which abundance scales (rare vs. abundant isoforms) are affected genome-wide.

2. **Monotonicity check:** By **Tsallis entropy** theory (Tsallis 2006), entropy is monotone decreasing in q . Divergence should NOT show erratic increases with q . Small fluctuations are normal, but large spikes indicate numerical instability.
3. **Comparison with genome-wide patterns:** Compute q -spectra for housekeeping genes (GAPDH, ACTB, etc.). These should show **BALANCED** patterns. If not, revisit normalization parameters.

S4 Workflow: Unified Analysis via TSENATAnalysis Objects

TSENAT provides an integrated S4-based workflow for coordinated analysis following three principles:

1. **Configure once:** Use `tsenat_config()` to specify analysis parameters (q -values, metadata columns)
2. **Run pipeline:** Use `tsenat()` to orchestrate all analysis steps in sequence
3. **Access results:** Use S4 accessor methods for type-safe result retrieval

The `tsenat()` function accepts a configured `TSENATConfig` object and orchestrates a complete pipeline: diversity computation $\rightarrow$ quality control $\rightarrow$ statistical testing $\rightarrow$ effect sizes (as specified by the `methods` parameter). Parameter `verbose = TRUE` outputs progress messages showing data filtering steps and method completion. This single-function approach ensures consistency between preprocessing and analysis steps, eliminating boilerplate and reducing risk of parameter mismatches.

Accessor methods return types: Most accessors return `SummarizedExperiment` objects (following Bioconductor conventions) containing results as assays and metadata:

- `diversity(analysis)` $\rightarrow$ `SummarizedExperiment` (genes $\times$ samples, multiple assays for each q -value)
- `lmResults(analysis)` $\rightarrow$ list containing `data.frame` of LM interaction results (or `NULL` if method not run)
- `jeoResults(analysis, q = 1.0)` $\rightarrow$ `SummarizedExperiment` of jackknife entropy outlier confidence intervals (or `NULL`)

- `jisResults(analysis, q = 1.0)` → list of jackknife isoform switching results (or NULL)
- `rankResults(analysis)` → data.frame of rank test results (or NULL)
- `divergence(analysis)` → SummarizedExperiment of pairwise divergence metrics

If a requested method was not run during `tseNat()` orchestration, the accessor returns NULL—users should check results before downstream use.

```
# Configure analysis parameters once (best practice for reproducibility)
tseNat_config <- tseNat_config(
  q_values = seq(0, 2, by = 0.05),
  sample_type_col = "sample_type",
  subject_col = "paired_samples"
)

# Run complete pipeline with one function call orchestrating all specified methods
# methods parameter: vector of analysis steps to execute ("diversity", "lm_interaction", "jackknife", "
# verbose = TRUE: prints progress messages showing gene filtering, step completion, and result summarie
analysis <- tseNat(
  se = se,
  config = tseNat_config,
  methods = c("diversity", "lm_interaction", "jackknife", "divergence"),
  verbose = TRUE
)

# Access results via consistent S4 methods (type-safe, mutually-exclusive access instead of nested list
diversity_results <- diversity(analysis)
lm_table <- lmResults(analysis)
jk_entropy <- jeoResults(analysis, q = 1.0)
jk_iso <- jisResults(analysis, q = 1.0)
rank_test <- rankResults(analysis)
divergence_table <- divergence(analysis)
```

For detailed step-by-step implementation, see the sections above demonstrating each analysis component and its accessor patterns.

Appendices

This main vignette is complemented by two comprehensive appendices:

Appendix A: Equivalence Validation — TSENAT vs SplicingFactory

Validates that TSENAT's Shannon and Simpson entropy implementations are mathematically equivalent to SplicingFactory. Includes:

- Mathematical proof of equivalence for $q=1$ (Shannon) and $q=2$ (Simpson)
- Benchmarking on TCGA BRCA RNA-seq data
- Detailed comparison tables and visualizations
- When to use TSENAT vs SplicingFactory

View Appendix A →

TSENAT Appendix B: Non-Parametric Validation of Linear Model Results via GAM and Rank-Based Methods

Objective: Validate q -value $\times$ group interaction detection results from linear models using complementary non-parametric statistical approaches.

TSENAT’s default linear modeling approach (via `calculate_difference_s4()`) provides parametric tests for detecting scale-dependent differences in Tsallis entropy across q -values and experimental groups. To ensure robustness and generalization, Appendix B presents two independent, non-parametric alternatives that serve as validation methods:

Approach 1: Generalized Additive Model (GAM) with ARIMA-Ordered Time Series - Treats q -values as explicit sequential measurements (ordered from low to high) - Applies ARIMA(1,1,0) differencing to account for autocorrelation between adjacent q -points - Fits smooth additive functions of q and group effects (non-parametric basis functions) - Automatically detects heteroscedasticity and applies optimal weighting - Reference: Goude (2024)

Approach 2: Rank-Based Friedman Test with Hochberg Correction - Non-parametric alternative requiring no distributional assumptions - Friedman test for detecting differences across multiple q -values within each group - Accounts for paired/blocked structure in q -ordered measurements - Hochberg step-up procedure for controlling family-wise error rate (FWER) - References: Efron & Tibshirani (1993); Benjamini & Hochberg (1995)

Both methods incorporate principled advances to handle real-world RNA-seq data: - **Heteroscedasticity detection and weighting:** Entropy variance often increases with q -value; both approaches adaptively weight measurements - **Boundary condition handling:** Low and high q -values exhibit different variance properties; explicit boundary adaptation improves reliability - **Automatic test selection:** Model diagnostics trigger appropriate method switches based on data structure

Appendix B provides: - Detailed statistical methodology for both approaches - Step-by-step implementation examples - Comparison of linear model results vs. GAM vs. rank-based findings - Concordance tables showing agreement across methods - Recommendations for when each method is most appropriate - Power analysis and sample size guidance for different designs

Key Finding: High concordance (typically >90% agreement) across methods validates that linear model discoveries are robust to model assumptions and generalize to non-parametric frameworks.

View Appendix B →

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Session Information

```
sessionInfo()
#> R version 4.5.2 (2025-10-31)
#> Platform: x86_64-conda-linux-gnu
#> Running under: Ubuntu 22.04.5 LTS
#>
#> Matrix products: default
#> BLAS/LAPACK: /home/nouser/miniconda3/lib/libopenblas-r0.3.30.so; LAPACK version 3.12.0
#>
#> locale:
#>  [1] LC_CTYPE=es_ES.UTF-8      LC_NUMERIC=C
#>  [3] LC_TIME=de_DE.UTF-8      LC_COLLATE=es_ES.UTF-8
#>  [5] LC_MONETARY=de_DE.UTF-8  LC_MESSAGES=es_ES.UTF-8
#>  [7] LC_PAPER=de_DE.UTF-8     LC_NAME=C
#>  [9] LC_ADDRESS=C             LC_TELEPHONE=C
#> [11] LC_MEASUREMENT=de_DE.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: Europe/Berlin
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats4      stats      graphics  grDevices  utils      datasets  methods
#> [8] base
#>
```



```

#> other attached packages:
#> [1] gridExtra_2.3                ggplot2_4.0.2
#> [3] cowplot_1.2.0                knitr_1.51
#> [5] dplyr_1.2.0                  SummarizedExperiment_1.40.0
#> [7] Biobase_2.70.0               GenomicRanges_1.62.1
#> [9] Seqinfo_1.0.0                IRanges_2.44.0
#> [11] S4Vectors_0.48.0            BiocGenerics_0.56.0
#> [13] generics_0.1.4              MatrixGenerics_1.22.0
#> [15] matrixStats_1.5.0           kableExtra_1.4.0
#> [17] TSENAT_0.99.0               testthat_3.3.2
#>
#> loaded via a namespace (and not attached):
#> [1] tidyselect_1.2.1            viridisLite_0.4.3          farver_2.1.2
#> [4] S7_0.2.1                    fastmap_1.2.0              digest_0.6.39
#> [7] lifecycle_1.0.5            magrittr_2.0.4             compiler_4.5.2
#> [10] rlang_1.1.7                 sass_0.4.10                tools_4.5.2
#> [13] yaml_2.3.12                 labeling_0.4.3             S4Arrays_1.10.1
#> [16] pkgbuild_1.4.8              DelayedArray_0.36.0        xml2_1.5.2
#> [19] RColorBrewer_1.1-3          pkgload_1.5.0              abind_1.4-8
#> [22] BiocParallel_1.44.0         withr_3.0.2                purrr_1.2.1
#> [25] geepack_1.3.13              desc_1.4.3                 grid_4.5.2
#> [28] scales_1.4.0                MASS_7.3-65                 tinytex_0.58
#> [31] cli_3.6.5                   rmarkdown_2.30             otel_0.2.0
#> [34] rstudioapi_0.18.0          tzdb_0.5.0                 cachem_1.1.0
#> [37] stringr_1.6.0               splines_4.5.2              parallel_4.5.2
#> [40] XVector_0.50.0              vctrs_0.7.2                Matrix_1.7-4
#> [43] jsonlite_2.0.0              hms_1.1.4                  systemfonts_1.3.2
#> [46] tidyr_1.3.2                 jquerylib_0.1.4            glue_1.8.0
#> [49] codetools_0.2-20           stringi_1.8.7              gtable_0.3.6
#> [52] tibble_3.3.1                pillar_1.11.1              htmltools_0.5.9
#> [55] brio_1.1.5                  R6_2.6.1                   textshaping_1.0.5
#> [58] rprojroot_2.1.1            evaluate_1.0.5             lattice_0.22-9
#> [61] readr_2.2.0                 backports_1.5.0            pheatmap_1.0.13
#> [64] memoise_2.0.1              broom_1.0.12               bslib_0.10.0
#> [67] Rcpp_1.1.1                  svglite_2.2.2              SparseArray_1.10.9
#> [70] nlme_3.1-168                mgcv_1.9-4                 xfun_0.56
#> [73] pkgconfig_2.0.3

```